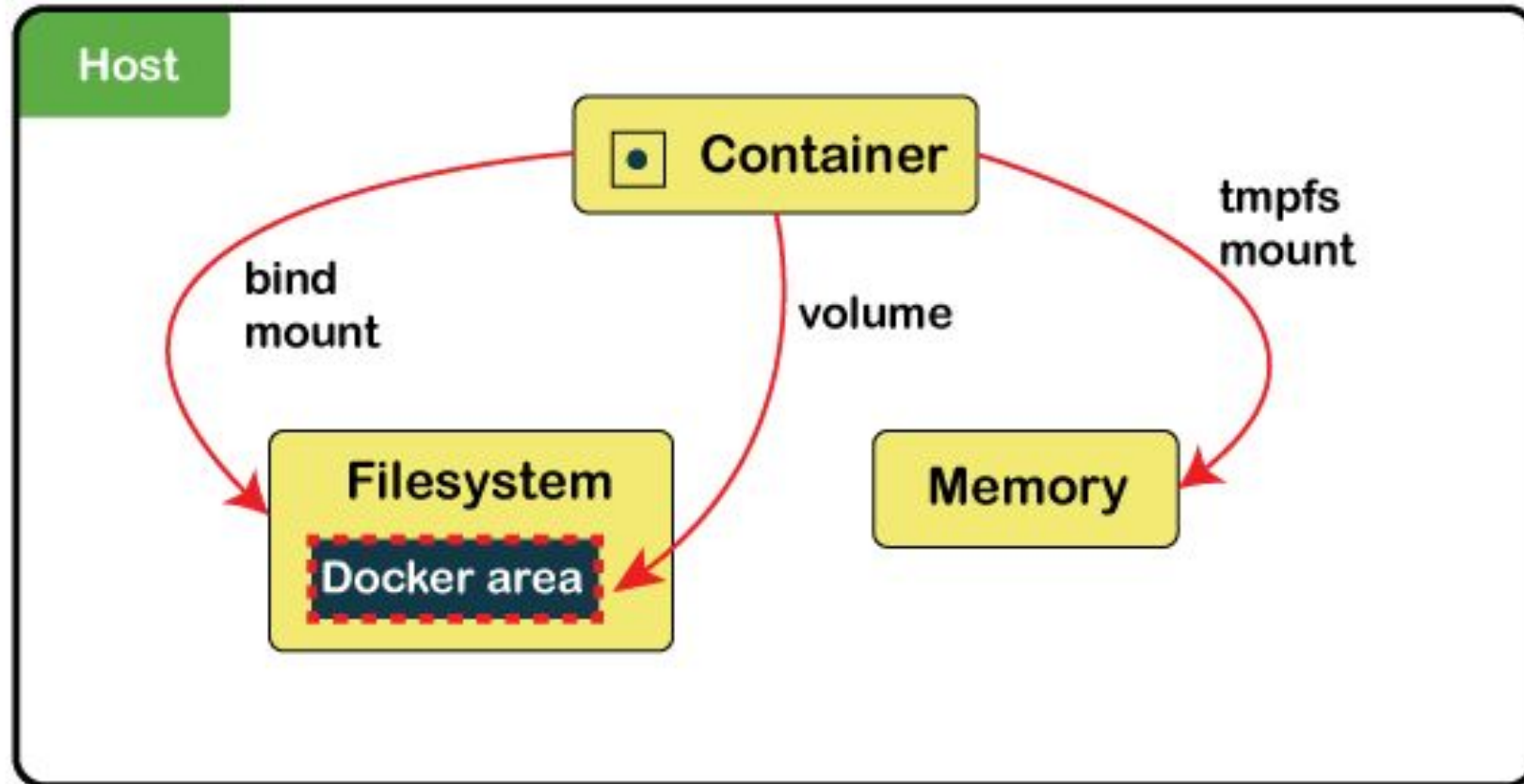




What is Docker Volume?



What is Docker Volume?

- Docker volumes are a widely used and useful tool for ensuring data persistence while working in containers. Docker volumes are file systems mounted on Docker containers to preserve data generated by the running container.
- The data doesn't persist when that container no longer exists, and it can be difficult to get the data out of the container if another process needs it.
- Docker has two options for containers to store files in the host machine so that the files are persisted even after the container stops:
 1. **Volumes**: Volumes are stored in a part of the host filesystem, managed by Docker (`/var/lib/docker/volumes/` on Linux).
 2. **Bind mounts**: Bind mounts may be stored anywhere on the host system.

Volume

- `$ docker volume create my-vol`
- `$ docker volume ls`
- `$ docker run -d -p 5432:5432 -e POSTGRES_PASSWORD=123 -v my-vol:/var/lib/postgresql/data postgres`

If volume not exist, it will be created automatically.

- `$ docker run -d -p 5433:5432 -e POSTGRES_PASSWORD=123 --mount source=my-vol,target=/app postgres`

target = destination; /app will be created if not exist

- `$ docker volume rm volume-name` # Remove volume by name first
- `$ docker volume prune` # Remove all unused volume

Volume

- `$ sudo vim Dockerfile`

```
FROM postgres:13.8  
  
VOLUME /var/lib/postgresql/data  
VOLUME /app  
  
ENV POSTGRES_PASSWORD=123  
ENV POSTGRES_DB=test  
ENV POSTGRES_USER=dara
```

- 1st **VOLUME** will be created automatically and mount to that path `“/var/lib/postgresql/data”`
- 2nd **VOLUME** will be created automatically and directory `“/app”` will be created if not exist and mount to it.

Volume

```
seng@ip-172-31-34-234:~$ docker run -d -p 5432:5432 -e POSTGRES_PASSWORD=123 pg-img
5e243976e38ad9718bc6a443e314c3e20ae897cc565806493e2870bbc387ad03
seng@ip-172-31-34-234:~$ docker volume ls
DRIVER      VOLUME NAME
local       45311d2e2398bb30a49ca6a7b5a13af3dfc9c3e5cdb2f9c2c8e1f6b01cf4f002
local       df69e58fa0d763800a9332bda5e98c700662a371bad85444f4a99d12bec5aa20
seng@ip-172-31-34-234:~$ docker exec -it 5e243976e38ad97 bash
root@5e243976e38a:/# ls
app  bin  boot  dev  docker-entrypoint-initdb.d  etc  home  lib  lib64  media  mnt  tmp  usr
root@5e243976e38a:/# cd /usr/lib/postgresql/
root@5e243976e38a:/usr/lib/postgresql# ls
15
```

9. Type of Mount

\$ docker run -d -p 2348:5432 --name my-postgres8 -e POSTGRES_PASSWORD=123
-v /app postgres:14 # add and mount /app to the generated volume

\$ docker run -d -p 2349:5432 --name my-postgres9 -e POSTGRES_PASSWORD=123
--volumes-from=my-postgres8 postgres:14 # mount volume from the existing
container

Bind mounts

- `$ docker run -d -p 5435:5432 -e POSTGRES_PASSWORD=123 -v /home/seng/pg-db:/var/lib/postgresql/data postgres`

If directory “pg-db” not exist, it will be created automatically.

- `$ sudo mkdir pg-db2 # pg-db2 must be existing before use`
- `$ docker run -d -p 5436:5432 -e POSTGRES_PASSWORD=123 --mount type=bind,source=/home/seng/pg-db2,target=/app postgres`

target = destination; /app will be created if not exist

Spring boot Dockerfile

```
FROM openjdk:11
ADD register-v2.jar ROOT.jar
EXPOSE 8080
ENTRYPOINT ["java", "-jar", "ROOT.jar"]
```

- `docker build -t register-img .`
- `docker run -d -p 8081:8080 -v /home/$USER/my-images:/src/main/resources/images register-img`

A large, faint watermark of the ISTAD logo is centered in the background. It is a circular emblem with a blue outer ring containing the text 'INSTITUTE OF SCIENCE AND TECHNOLOGY ADVANCED DEVELOPMENT' and Khmer text. Inside the ring is a blue circle with a white atomic symbol. The word 'ISTAD' is written in red across the bottom of the inner circle. Binary code '01010100 01000001 01000000' is visible along the top inner edge of the ring.

Thank you

Begin with the theory, Start with the practical